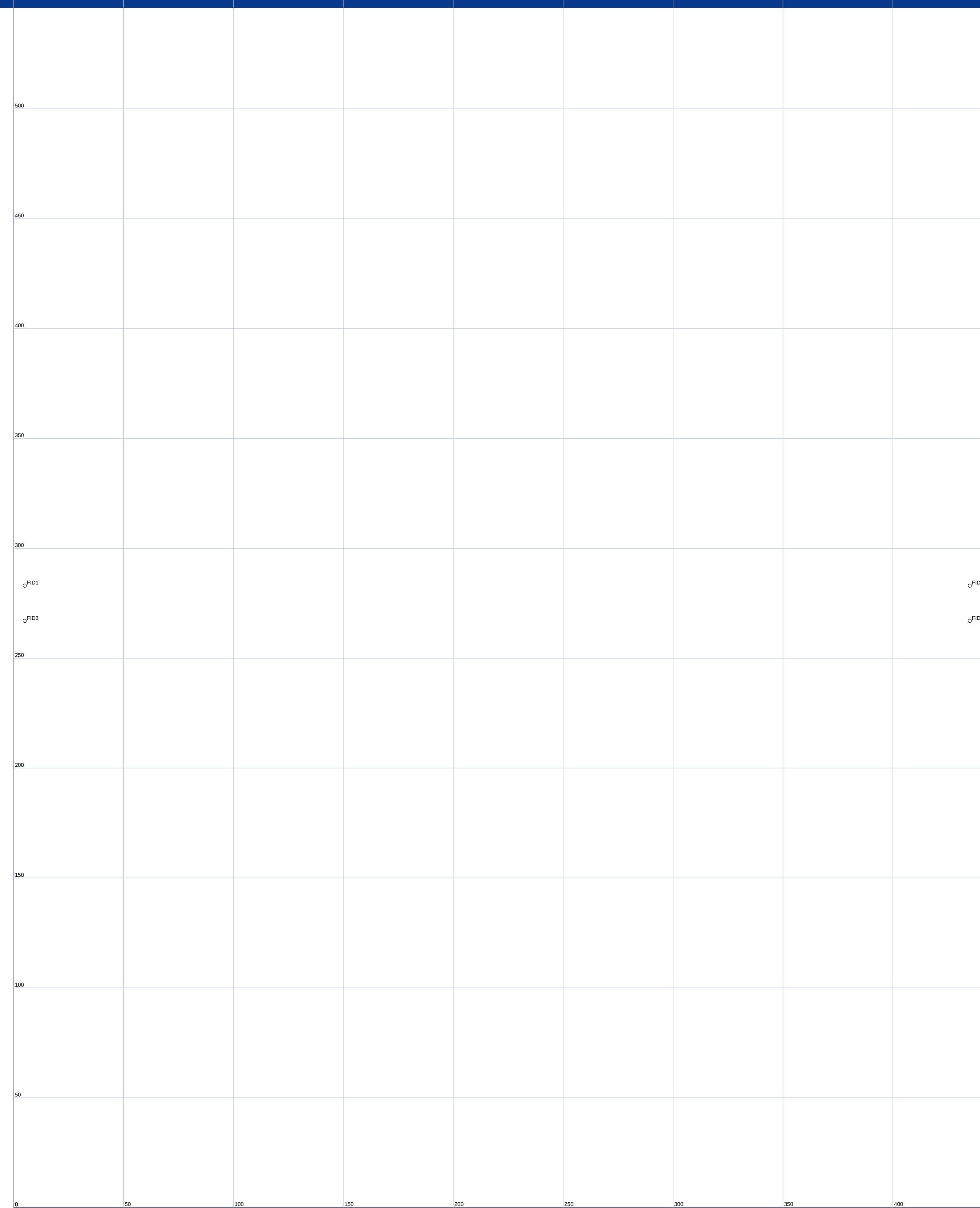


LR-S1A 1U CPO Switch Motherboard — Fabrication Drawing



Stackup — 18 layers · 3.20 mm

Layer	Thk	Material / role
Mask T	—	LPI green, Class T
L1 F.Cu	1.4 mil	4 oz signal · 100/50 Ω
PP 1	3.0 mil	Megtron-7 Dk 3.40
L2 In1	0.7 mil	2 oz GND ref
Core 1	4.0 mil	Megtron-7
L3 In2	0.7 mil	2 oz signal (HBM4)
PP 2	3.0 mil	Megtron-7
L4 In3	0.7 mil	2 oz GND ref
Core 2	4.0 mil	Megtron-7
L5 In4	0.7 mil	V_ASIC 0V85 plane
PP 3	3.0 mil	Faradflex BC24
L6 In5	0.7 mil	2 oz GND ref
Core 3	4.0 mil	Megtron-7
L7 In6	0.7 mil	2 oz signal CPO
PP 4	3.0 mil	Megtron-7
L8 In7	0.7 mil	2 oz GND ref
Core 4	5.0 mil	IT-150DA centre
L9 In8	0.7 mil	GND core ref
PP 5	3.0 mil	Megtron-7
L10 In9	0.7 mil	2 oz GND ref
Core 5	4.0 mil	Megtron-7
L11 In10	0.7 mil	V_OPT 1V2 plane
PP 6	3.0 mil	Faradflex BC24
L12 In11	0.7 mil	V_AUX 3V3 plane
Core 6	4.0 mil	Megtron-7
L13 In12	0.7 mil	2 oz GND ref
PP 7	3.0 mil	Megtron-7
L14 In13	0.7 mil	PCIe Gen6 ×16
Core 7	4.0 mil	Megtron-7
L15 In14	0.7 mil	2 oz GND ref
PP 8	3.0 mil	Megtron-7
L16 In15	0.7 mil	2 oz mgmt + OOB
Core 8	4.0 mil	Megtron-7
L17 In16	0.7 mil	2 oz GND ref
PP 9	3.0 mil	Megtron-7
L18 B.Cu	1.4 mil	4 oz + DrMOS phases
Mask B	—	LPI green, Class T

Drill chart (Excellon 2.0, mm)

Size	Type	Description	Qty
0.10	PTH	Microvia L1→L2 / L17→L18	12 480
0.20	PTH	Microvia stack L1→L3	6 200
0.25	PTH	Buried via In1↔In16	10 480
0.30	PTH	Standard sig (back-drill)	18 640
0.40	PTH	HBM4 strobe via	2 048
0.50	PTH	DrMOS power tap	1 920
0.60	PTH	BGA escape (LR-CSW)	484
0.80	PTH	DC power stitching	864
1.00	NPTH	Connector mount	64
2.20	NPTH	Fiducial	4
2.70	NPTH	Mounting hole M2.5	8

General fabrication notes

- Fabricate per IPC-6012 Class 3/A; accept per IPC-A-600 Class 3.
- Megtron-7 RF signal cores · IT-150DA centre core · Faradflex BC24.
- Cu 4 oz outer / 2 oz inner · ENEPIG 6 μin Au / 200 μin Pd / 150 μin Ni.
- Solder mask LPI green both sides · legend white epoxy both sides.
- Impedance 100 Ω diff ±5 % · 50 Ω SE ±5 % · coupon per IPC-2141.
- Back-drill PCIe-Gen6 / HBM4 nets to stub ≤ 6 mil (Polar Si9000).
- Drill-to-Cu ≥ 8 mil internal · Cu-to-edge ≥ 20 mil all layers.
- UL94 V-0 mark + date code on F.Silk (see Assembly Drawing).
- μvia stack filled-and-capped per IPC-4761 Type VII · 3 sub-stacks.
- CAF 1 000 h 85/85 50 V · IPC-TM-650 2.6.25 · R > 100 MΩ pass.
- Outline routed + chamfered R3.0 corners · edge plating optional.
- Panel 600×750 mm 1-up · mouse-bite 4×1.5 mm · 100 % bare-board ET.
- AOI + TDR every panel · cross-section first-article only.
- Vendor logo + date code on F.Silk 1 mm chars · bottom-left location.
- Vacuum-sealed pack · desiccant + HIC · MSL-3 J-STD-033.
- RoHS / REACH · Pb-free · ENEPIG 6 μin · no flash gold.
- Layer-to-layer registration ≤ 1.5 mil · GND-flanked signals.
- CTI ≥ 600 V · CTE-Z ≤ 50 ppm/°C below Tg · Tg ≥ 200 °C.

LightRail AI — Confidential / Investor data-room	
Programme	LR-S1A 1U CPO Switch Motherboard
Drawing No.	LR-S1A-FAB-001
Title	Fab Drawing — Sheet 1 of 1
Material	18-layer HDI · Megtron-7 / FR-4 / BC24
Designer	LightRail AI — Design Automation
Date / Rev	2026-04-19 · Rev 1.0